In [9]:
```
#Write a program to print your name, age, and college name on separate lines
print("Nandini Naresh Battul")
print(21)
```

```
Nandini Naresh Battul
21
```

In [15]:
```
#Take two numbers as input and print their addition, subtraction, multiplication, a
a = int(input("a = "))
b = int(input("b = "))
print("Addition:- ",a+b)
print("Subtraction:- ",a-b)
print("Multiplication:- ",a*b)
print("Division:- ",a//b)
```

```
Addition:-  120
Subtraction:-  54
Multiplication:-  2871
Division:-  2
```

In [19]:
```
#Take an integer as input and check whether it is even or odd.
num = int(input("Enter a Number: "))
if num%2 == 0:
    print(num,"is Even")
else:
    print(num,"is Odd")
```

```
8 is Even
```

In [23]:
```
#Take three numbers as input and print the largest number using if-elif-else.
a = int(input("a = "))
b = int(input("b = "))
c = int(input("c = "))
if a>b and a > c:
    print(a,"is greater")
elif b>c and b>a:
    print(b,"is greater")
else:
    print(c,"is greater")
```

```
9 is greater
```

In [25]:
```
#Take a number as input and check whether it is positive, negative, or zero.
num = int(input("Enter num: "))
if num > 0:
    print(num,"is Positive")
elif num < 0:
    print(num,"is Negative")
else:
    print(num,"is Zero")
```

```
8 is Positive
```

In [45]:
```
#Take a number as input and print its multiplication table from 1 to 10 using a for
num = int(input("Enter num: "))
```

```
for i in range(1,11):
    print(f"{num} x {i} = {num*i}")
```

```
6 x 1 = 6
6 x 2 = 12
6 x 3 = 18
6 x 4 = 24
6 x 5 = 30
6 x 6 = 36
6 x 7 = 42
6 x 8 = 48
6 x 9 = 54
6 x 10 = 60
```

In [55]:
```
#Write a program to calculate the sum of numbers from 1 to n, where n is entered by
n = int(input("Enter num: "))
sum = 0
for i in range(1,n+1):
    sum = sum + i
    i+=1
print("Sum is ",sum)
```

```
Sum is  55
```

In [59]:
```
#Take a string as input and print it in reverse order
s = str(input("Enter String: "))
rev = s[::-1]
print("The Reverse String of",s,"is",rev)
```

```
The Reverse String of Samarth is htramaS
```

In [71]:
```
#Take a string as input and count how many vowels (a, e, i, o, u) it contains
s = str(input("Enter String: "))
vowels = 'aeiouAEIOU'
count = 0
for i in s:
    if i in vowels:
        count = count + 1
print(count)
```

```
4
```

In [75]:
```
#Given the list [12, 45, 23, 67, 34], write a program to find the largest number wi
l1 = [12,45,23,67,34]
max_num = max(l1)
print(max_num)
```

```
67
```

In [81]:
```
#Write a Python program to calculate the factorial of a number using a for loop.
n = int(input("Enter num: "))
fact = 1
for i in range(n,1,-1):
    fact = fact * i
print("Fcatorial is ",fact)
```

```
Fcatorial is  120
```

In [97]:
```
#Write a program to check whether a number entered by the user is prime or not.
num = int(input("Enter a Number: "))
for i in range(2,num):
    if num%i == 0:
        print(num,"is not Prime Number")
    else:
        print(num,"is a Prime Number")
```

```
3 is a Prime Number
```

In [ ]:
```
#Write a program to count the number of digits in an integer using a while loop.
```

In [115…
```
#Check For Perfect Number
def perfect_num(n):
    if n <= 1:
        return False
    sum = 0
    for i in range(1,n):
        if n%i==0:
            sum+=i
    return sum == n
print(perfect_num(28))
```

```
True
```

In [ ]:
```
#Count Digit In a Number
def count_digit(n):

```

In [7]:
```
num = int(input("Enter a Number: "))
count = 0
for i in range(2,num):
    if num%i == 0:
        count+=1
        i+=1
print(count)
```

```
14
```

In [11]:
```
name = input("Enter your Name ")
age = int(input("Enter age "))
print("After 20 years you are at ",age+20)
print(type(name))
print(type(age))
```

```
After 20 years you are at  40
<class 'str'>
<class 'int'>
```

In [13]:
```
marks = [89,76,87,69,73,78]
average = sum(marks) / len(marks)
print("Average Score: ",average)
```

```
Average Score:  78.66666666666667
```

In [17]:
```
name = "Nandini"
subject = "Python"
```

```
message = f"Hello {name}, Welcome to {subject} class!"
print(message)
```

```
Hello Nandini, Welcome to Python class!
```

In [19]:
```
# sequence data
students = ["Aditi", "Rahul", "Sneha"] # list
info = (101, "2005-06-15", "MCA01") # tuple
subjects = {"Python", "ADBMS", "NT"} # set
print("Students:", students)
print("Immutable Info:", info)
print("Unique Subjects:", subjects)
```

```
Students: ['Aditi', 'Rahul', 'Sneha']
Immutable Info: (101, '2005-06-15', 'MCA01')
Unique Subjects: {'ADBMS', 'NT', 'Python'}
```

In [21]:
```
#while loop
print("Output of while loop")
count = 0
while count < 5:
    print(count)
    count += 1
```

```
Output of while loop
0
1
2
3
4
```

In [31]:
```
numbers = [10,20,30,40]
print(numbers[0])
print(numbers[-2])
numbers.append("Apple")
print(numbers)
numbers.insert(2,"Pink")
print(numbers)
numbers.remove(30)
print(numbers)
numbers.pop()
print(numbers)
```

```
10
30
[10, 20, 30, 40, 'Apple']
[10, 20, 'Pink', 30, 40, 'Apple']
[10, 20, 'Pink', 40, 'Apple']
[10, 20, 'Pink', 40]
```

In [33]:
```
#Length
numbers = [1, 2, 3, 4, 5, 7,5,9]
print("No. of items in list are ",len(numbers))
# Sum
print("Sum of the list numbers is ", sum(numbers))
# sorting
```

```
print("List in Ascending order ",sorted(numbers))
print("List in descending order ",sorted(numbers, reverse=True))
```

```
No. of items in list are  8
Sum of the list numbers is  36
List in Ascending order  [1, 2, 3, 4, 5, 5, 7, 9]
List in descending order  [9, 7, 5, 5, 4, 3, 2, 1]
```

In [35]:
```
f= open("py_sample.txt")
print(f.read())
f.close()
```

```
I am Nandini Naresh Battul
I am Studying MCA
```

In [37]:
```
with open("py_sample.txt") as f:
 print(f.read(6))
 ct = f.read()
 ws = ct.split()
print("Total characters :- ",len(ct))
print("Total Words ",len(ws))
```

```
I am N
Total characters :-  39
Total Words  7
```

In [ ]: